TECHNICAL ASSESSMENT SUBMISSION

DETECT TECHNOLOGIES | TAP-JOB-ID-2507

Candidate Name:	Suprit Lenkennavar
Institute:	TAP Academy
Deadline:	Thursday, 28-05-2026, 11:00 AM
Status:	**All 3 Problems Fully Completed & Verified**

Submission Links

GitHub Repository: Contains all version-controlled source code, requirements, and documentation.	**GitHub Repo Link** **https://github.com/suprit-4618/...Assignment.git**
Google Drive Folder: Contains the raw high-resolution input and output demo videos (bypassing the Form's 10MB limit).	**Google Drive Link** **https://drive.google.com/drive/folders/1vQ6xd...**

Interactive Review (Google Colab)

To make evaluation as simple as possible, the following interactive Google Colab notebooks have been pre-configured. The reviewer can run the pipelines on GPU with a single click:

Problem 2: Face Blur	**Open Face Blur Notebook**
Problem 3: License Plates	**Open Plate Detection Notebook**

Summary of Completed Assignments

1. Move Zeros to the Right (Easy)
• Implementation: O(n) Time / O(n) Space algorithm keeping non-zero relative order.
• Location: problem1_move_zeros/solution.py
• Verification: Passed all 10 edge-case test suites (negative numbers, empty arrays, etc.).

2. Face Blur on Video (Medium)
• Implementation: OpenCV Haar Cascade detectors (Frontal + Profile) with IoU deduplication and dynamic Gaussian blur padding.
• Location: problem2_face_blur/face_blur.py
• Output: 12,642 verified real-time blurred face detections across 1,080 frames of street crowd footage.

3. License Plate Detection & OCR (Hard)
• Implementation: Haar cascade + contour aspect-ratio filtering, four-point perspective correction, and EasyOCR engine.
• Location: problem3_number_plate_detection/number_plate_detector.py
• Optimization: CPU skip-frame processing engine (runs OCR every N frames; copies logs on intermediate frames).
• Output: Extremely high-accuracy reads (e.g., N28-800928, confidence 1.00) logged in sample_output.json.